

HealthConnect Clinic

Initial Data Analysis Document

Week 4 – Project Kickoff & Problem Understanding | Data Analytics Track

1. Dataset Overview

The HealthConnect Appointment Dataset contains 5,000 appointment records and 18 variables. The dataset provides information relating to patient demographics, appointment scheduling, previous appointment history, reminder activity, accessibility factors and appointment outcomes.

The dataset includes variables such as patient age and gender, appointment type, booking and appointment dates, appointment day and time, booking lead time, previous appointments and no-shows, reminder information, distance to the clinic, estimated waiting time and final appointment outcome.

The dataset will be used to explore appointment attendance and no-show patterns and to identify factors that may be associated with missed appointments.

2. Problem Context

HealthConnect Clinic aims to better understand missed appointments and the factors that may influence whether patients attend or miss scheduled appointments. Missed appointments can affect clinic efficiency, appointment-slot utilisation and patient service delivery.

The Data Analytics component of the project will focus on analysing appointment records to identify attendance and no-show patterns. The analysis will investigate potential factors associated with missed appointments, including appointment characteristics, previous appointment behaviour, reminders, waiting time and distance to the clinic.

The findings will provide an analytical foundation for identifying potential opportunities to reduce missed appointments and improve the overall patient support experience.

3. Data Quality Assessment

An initial data-quality assessment was conducted in Power Query to evaluate the dataset's completeness, uniqueness, data types, consistency and potential unusual values.

Data Quality Area	Finding	Assessment
Duplicate records	No duplicate records found	No duplicate issue identified
Data types	All 18 variables match the Data Dictionary	No data-type issue identified
Inconsistent/invalid values	No inconsistent or invalid values identified during the initial review	No issue identified at this stage

Missing values	distance_to_clinic_km has 90 missing values (2%)	Requires consideration during later analysis
Missing values	waiting_time_minutes has 60 missing values (1%)	Requires consideration during later analysis
Unusual values/outliers	No obvious unusual values identified from the initial minimum, maximum and average review	Further exploratory analysis may provide additional insight

Numeric Variable Review

Variable	Minimum	Maximum	Average
Age	18	80	48.80
Booking Lead Days	0	60	29.64
Previous Appointments	0	11	3.01
Previous No-Shows	0	5	0.54
Distance to Clinic (km)	0.5	45	10.11
Waiting Time (minutes)	2	68	24.19

Initial conclusion: The dataset appears suitable for further exploratory analysis. However, the missing values in distance_to_clinic_km and waiting_time_minutes should be considered when analysing relationships involving those variables.

4. Relevant Variables for Analysis

Outcome Variable

- appointment_outcome — the primary outcome variable used to measure appointment attendance and identify no-show patterns.

Patient Demographic Variables

- gender
- age
- age_group

These variables may help determine whether appointment attendance patterns differ across demographic groups.

Appointment and Scheduling Variables

- appointment_type
- booking_date
- appointment_date

- appointment_day
- appointment_time
- booking_lead_days

These variables will support analysis of whether scheduling characteristics, appointment timing and the time between booking and the appointment are associated with attendance outcomes.

Previous Appointment History

- previous_appointments
- previous_no_shows

These variables are important for investigating whether a patient's previous appointment behaviour is associated with future attendance or no-show patterns.

Reminder Variables

- reminder_sent
- reminder_channel

These variables will support an initial assessment of whether appointment reminders and reminder channels are associated with appointment attendance.

Accessibility and Service Variables

- distance_to_clinic_km
- waiting_time_minutes

These variables may help investigate whether accessibility and waiting time are associated with missed appointments.

Identifier Variables

- appointment_id
- patient_id

These variables will primarily be used to identify and count appointment records and patients. appointment_id can support checks for unique appointments, while patient_id may support analysis of repeat appointment behaviour.

5. Relevant Business Questions

1. What is the overall appointment attendance and no-show rate?

This will provide a baseline understanding of the scale of missed appointments at HealthConnect Clinic.

2. Which appointment and scheduling factors are associated with higher no-show rates?

The analysis will investigate factors such as appointment type, appointment day, appointment time and booking lead days.

3. Does previous appointment behaviour appear to be associated with future no-shows?

The analysis will investigate whether patients with previous no-shows are more likely to miss subsequent appointments.

4. Are appointment reminders associated with improved attendance?

The analysis will compare appointment outcomes based on whether a reminder was sent and, where relevant, the reminder channel used.

5. Do accessibility and service-related factors appear to influence attendance?

The analysis will investigate whether distance to the clinic and estimated waiting time are associated with different appointment outcomes.

6. Do no-show patterns differ across patient demographic groups?

The analysis will examine attendance patterns across age groups and gender categories.

These questions are intended to guide exploratory analysis rather than establish causal relationships. The analysis can identify patterns and associations in the available data, but further investigation may be required to determine the underlying causes of those patterns.

6. Proposed Key Performance Indicators (KPIs)

KPI	Description	Linked Business Question
Appointment Attendance Rate	Percentage of scheduled appointments that result in attendance.	What is the overall appointment attendance rate?
No-Show Rate	Percentage of scheduled appointments that result in a no-show.	What is the overall no-show rate, and which groups have higher rates?
Reminder Coverage Rate	Percentage of appointments for which a reminder was sent.	Are reminders being used consistently, and are they associated with attendance?
Repeat No-Show Rate	Proportion of appointments involving patients with a previous history of no-shows that result in another no-show.	Does previous appointment behaviour appear to be associated with future no-shows?
Average Booking Lead Time	Average number of days between booking and the scheduled appointment.	Is a longer or shorter booking lead time associated with attendance patterns?

These KPIs will provide a starting point for monitoring appointment performance and investigating factors associated with missed appointments. Their final calculation and use may be refined after exploratory analysis of the dataset.

7. Initial Analysis Approach

Step 1: Review the Dataset and Data Dictionary

The dataset structure, variable definitions and data types will be reviewed to ensure that each variable is correctly understood and used appropriately during the analysis.

Step 2: Assess Data Quality

The dataset will be reviewed for missing values, duplicate records, incorrect data types, inconsistent or invalid values, and unusual values. Any identified issues will be documented before further analysis.

Step 3: Prepare the Data for Analysis

Where necessary, data-quality issues will be addressed. The missing values in `distance_to_clinic_km` and `waiting_time_minutes` will be considered carefully before deciding on an appropriate treatment. Records will not be removed without assessing the potential impact on the analysis.

Step 4: Establish Baseline Appointment Performance

The overall distribution of `appointment_outcome` will be analysed to establish baseline appointment attendance and no-show rates.

Step 5: Explore No-Show Patterns

No-show patterns will be compared across relevant demographic, appointment, history, reminder, accessibility and waiting-time variables.

Step 6: Investigate Potential Associations

The analysis will examine whether particular characteristics or factors are associated with higher or lower no-show rates through comparisons across relevant groups and identification of notable patterns.

Step 7: Develop KPIs and Visualisations

Following exploratory analysis, the proposed KPIs will be calculated and refined where necessary. Appropriate visualisations and dashboard components may then be developed to communicate key findings.

Step 8: Interpret Findings and Identify Opportunities

The final stages of the analysis will focus on identifying the most important patterns, potential risk factors and opportunities for reducing missed appointments and improving appointment-slot utilisation.

8. Assumptions, Limitations and Key Considerations

Assumptions

- The appointment records are assumed to be accurately recorded and representative of HealthConnect Clinic's appointment activity.
- The values in appointment_outcome are assumed to accurately reflect the final status of each appointment.
- The Data Dictionary is assumed to provide the correct definitions and intended interpretation of each variable.
- The variables included in the dataset are assumed to be sufficient for conducting an initial exploratory analysis of appointment attendance and no-show patterns.

Limitations

- The available data can identify patterns and associations, but it may not establish that a particular factor directly causes a patient to miss an appointment.
- The dataset may not include all factors that influence appointment attendance.
- Because the dataset contains anonymised records, the analysis is limited to the variables provided and cannot incorporate additional individual-level information.
- The dataset contains 90 missing values (2%) in distance_to_clinic_km and 60 missing values (1%) in waiting_time_minutes. Analyses involving these variables may therefore be based on fewer records.

Key Risks and Dependencies

- The treatment of missing values may affect analyses involving distance to the clinic and waiting time.
- The interpretation of no-show patterns depends on the categories and definitions contained in appointment_outcome.
- Further analysis will depend on confirming how appointment outcomes should be grouped for KPI calculations, particularly for defining attendance and no-show rates.
- Any future dashboard or advanced analysis will depend on maintaining consistent data definitions and calculation methods.